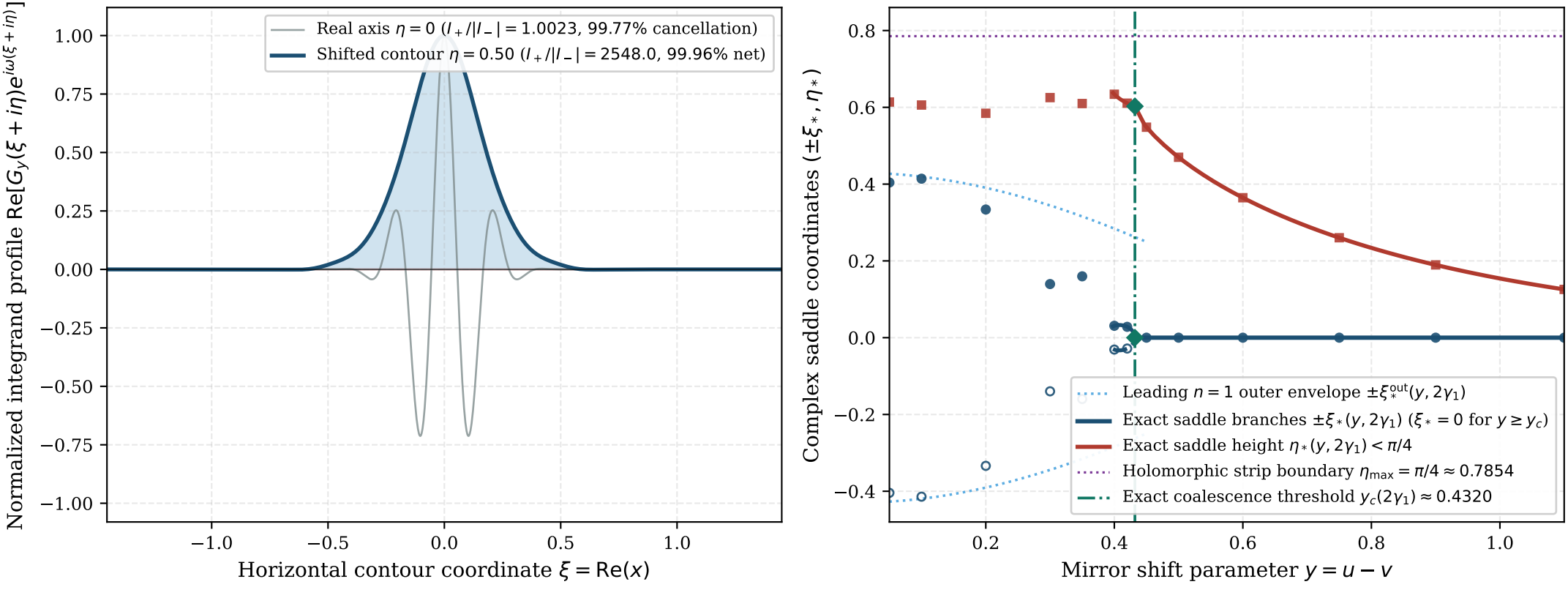

(a) Elimination of Real-Axis Cancellation ($y = 0.40, \omega = 2\gamma_1$)
Real axis $\eta = 0$ ($I_+/|I_-| = 1.0023$, 99.77% cancellation)
Shifted contour $\eta = 0.50$ ($I_+/|I_-| = 2548.0$, 99.96% net)
Normalized integrand profile $\mathrm{Re}[G_y(\xi + i\eta)e^{i\omega(\xi+i\eta)}]$
Horizontal contour coordinate $\xi = \mathrm{Re}(x)$
(b) Pitchfork Saddle-Point Coalescence at $\omega = 2\gamma_1$
Complex saddle coordinates ($\pm\xi_*, \eta_*$)
Leading $n = 1$ outer envelope $\pm\xi_*^{\mathrm{out}}(y, 2\gamma_1)$
Exact saddle branches $\pm\xi_*(y, 2\gamma_1)$ ($\xi_* = 0$ for $y \geq y_c$)
Exact saddle height $\eta_*(y, 2\gamma_1) < \pi/4$
Holomorphic strip boundary $\eta_{\max} = \pi/4 \approx 0.7854$
Exact coalescence threshold $y_c(2\gamma_1) \approx 0.4320$
Mirror shift parameter $y = u - v$